Ministry of Higher Education and Research
- Sidi Bel Abbes

Second Year Second Cycle - Artificial Intelligence and Data Science

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Manual of Realization: Malicious URL Detection Model

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Abstract

This project develops an AI-based model to detect malicious URLs using the Malicious URLs dataset. The model employs natural language processing (NLP) techniques, including TF-IDF and BERT, alongside machine learning (Random Forest) and deep learning (LSTM) to classify URLs into benign, phishing, malware, or defacement categories. Performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, with Random Forest achieving superior results.

0.1 Introduction

This project aims to develop an artificial intelligence (AI)-based model to detect malicious Uniform Resource Locators (URLs) using the Malicious URLs dataset. Malicious URLs are commonly used in cyberattacks, such as phishing and malware distribution. The proposed model leverages NLP techniques (TF-IDF and BERT embeddings), machine learning (Random Forest), and deep learning (LSTM) to classify URLs into four categories: benign, phishing, malware, and defacement. The methodology includes feature extraction, feature selection, model training, and performance evaluation to ensure high accuracy and low false positive rates.

0.2 Methodology

0.2.1 Dataset

The dataset is sourced from Kaggle, containing URLs labeled as benign, phishing, malware, or defacement. Preprocessing steps included removing missing values, cleaning URLs (e.g., removing protocols and special characters), and encoding labels for multi-class classification.

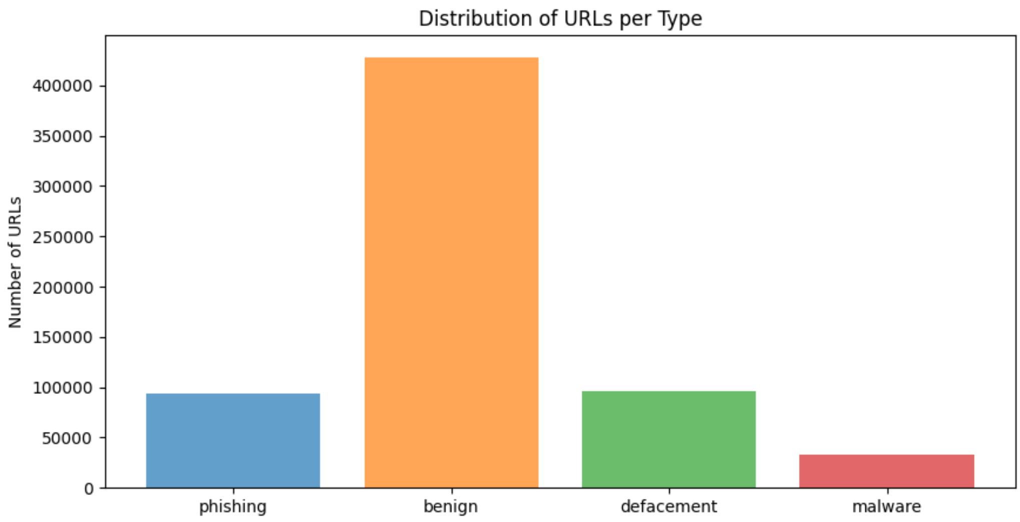


Figure 1: Distribution of URLs per Type

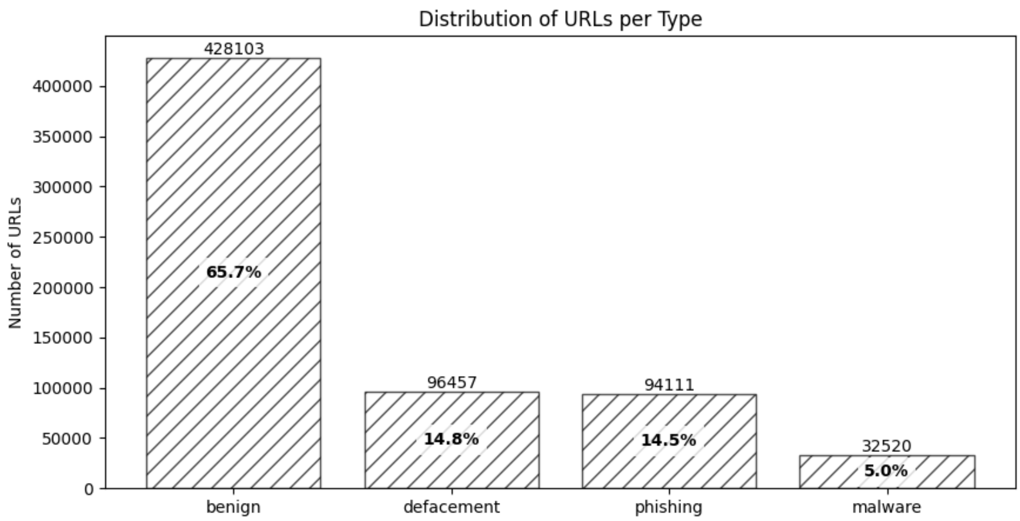


Figure 2: Distribution of URLs per Type with Percentage

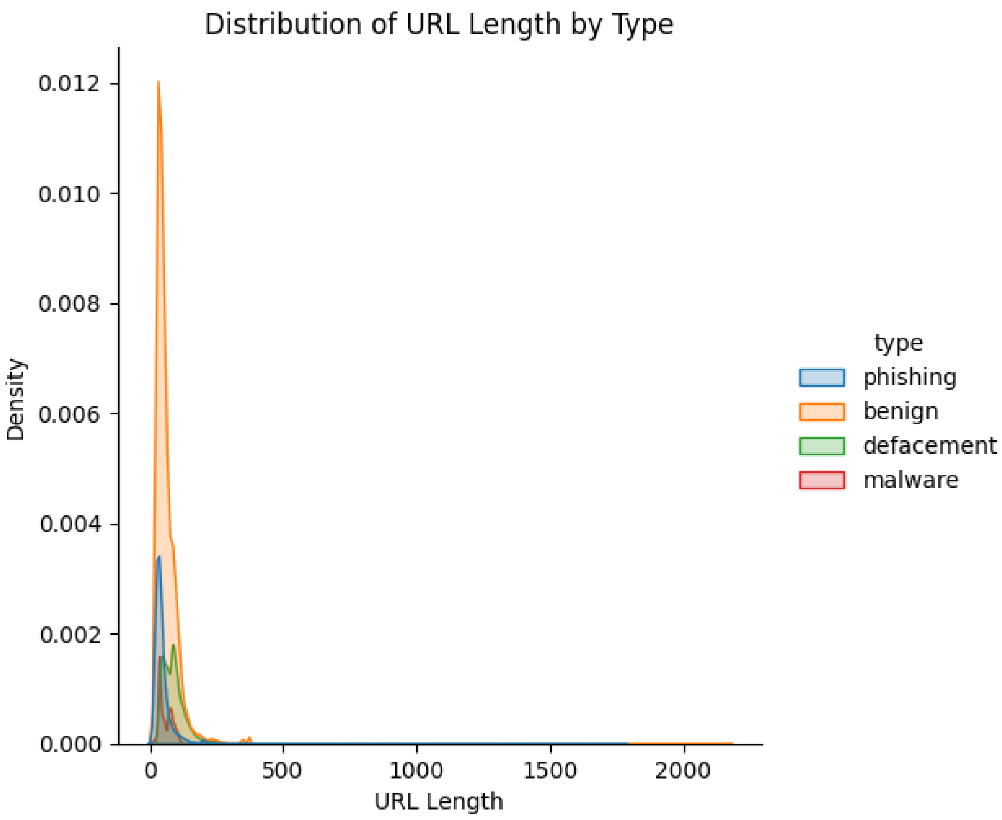


Figure 3: Distribution of URL Length by Type

0.2.2 Feature Extraction

Two feature extraction methods were employed:

- **TF-IDF**: Character-level n-grams (2–4) were extracted, resulting in a sparse matrix with 1000 features to reduce memory usage.
- **BERT**: The `bert-base-uncased` model generated 768-dimensional embeddings, limited to 10,000 samples due to computational constraints.

Feature selection used Principal Component Analysis (PCA) to reduce TF-IDF features to 100 components.

0.2.3 Models

Three models were trained sequentially:

- **Logistic Regression (LR)**: Trained on URL-specific features with hyperparameter tuning via Grid Search (parameters: C, solver).
- **LSTM**: A deep learning model with 128 LSTM units, two dense layers (64 units and 4 units with softmax activation), and dropout (0.5) to prevent overfitting. It was trained on URL-specific features reshaped into sequences, using categorical cross-entropy loss and the Adam optimizer (learning rate: 0.001) for 10 epochs with a batch size of 128.
- **LSTM with TF-IDF**: An LSTM model with the same architecture, trained on TF-IDF features. The TF-IDF matrix was reshaped into 3D tensors to match LSTM input requirements. This model was selected for deployment due to its superior performance in capturing textual patterns.

0.2.4 Evaluation

Models were evaluated using accuracy, precision, recall, and F1-score, computed on the test set. For LSTM models, training dynamics were visualized using accuracy and loss curves over epochs. Confusion matrices were generated to analyze class-specific performance, particularly for the imbalanced malware class. ROC-AUC curves were computed for Logistic Regression to assess discriminative ability.

0.3 Implementation

The project was implemented in Python 3.12 using a robust tech stack:

- **Libraries**: `scikit-learn` for Logistic Regression and TF-IDF, `tensorflow` for LSTM models, `urllib` and `tld` for URL parsing, and `gradio` for the web interface.

- **Notebook**:(`Data`$_{S}ecurity_{M}ini_{P}roject.ipynb)containsallpreprocessing, featureextraction, modeltraining, evaluation, andv$ $TheLSTMwithTF-IDFmodelwassavedas$ The Gradio interface was optimized for low latency, ensuring practical deployment in real-world scenarios.

0.4 Results

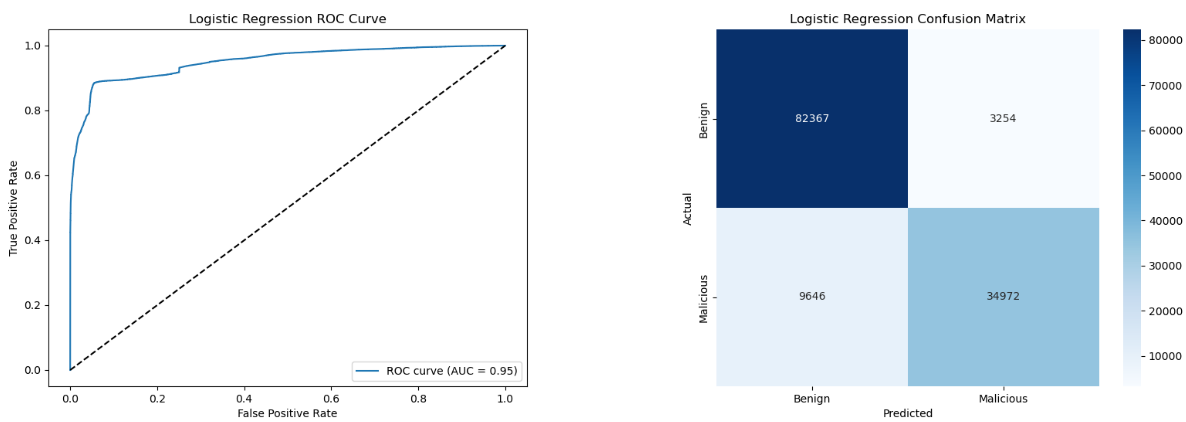
Performance metrics for Logistic Regression, LSTM, and LSTM with TF-IDF are presented in Tables 1, 2, and 3. Visualizations, including ROC curves, confusion matrices, and accuracy/loss curves, provide insights into model performance.

0.4.1 Logistic Regression

Logistic Regression achieved high accuracy (0.96) with balanced performance across classes, as shown in Table 1. The ROC curve and confusion matrix (Figure 4) highlight its ability to distinguish benign URLs but indicate slightly lower recall for phishing URLs.

Class	Precision	Recall	F1-Score	Support
Benign	0.97	0.99	0.98	85621
Defacement	0.98	0.99	0.99	19292
Malware	0.98	0.93	0.96	6504
Phishing	0.91	0.85	0.88	18822
Accuracy		0.96		130239
Macro Avg	0.96	0.94	0.95	130239
Weighted Avg	0.96	0.96	0.96	130239

Table 1: Logistic Regression Classification Report



(a) LR ROC Curve

(b) LR Confusion Matrix

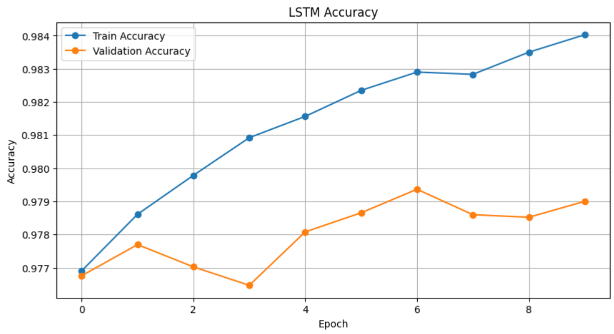
Figure 4: Logistic Regression Performance Visualizations

0.4.2 LSTM

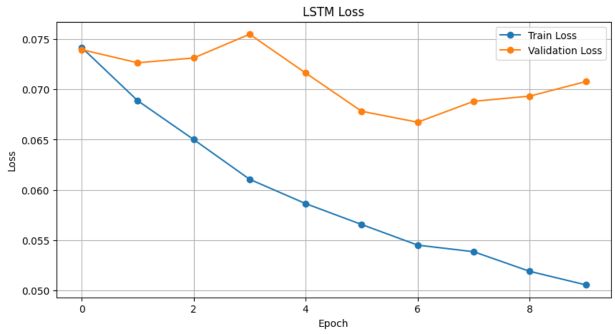
The LSTM model, trained on URL-specific features, achieved an accuracy of 0.9691, with strong precision (0.9723), recall (0.9363), and F1-score (0.9540), as shown in Table 2. The accuracy and loss curves (Figure 5) indicate stable convergence, while the confusion matrix reveals high performance across all classes, with minor misclassifications for malware URLs.

Metric	Value
Accuracy	0.9691
Precision	0.9723
Recall	0.9363
F1-Score	0.9540

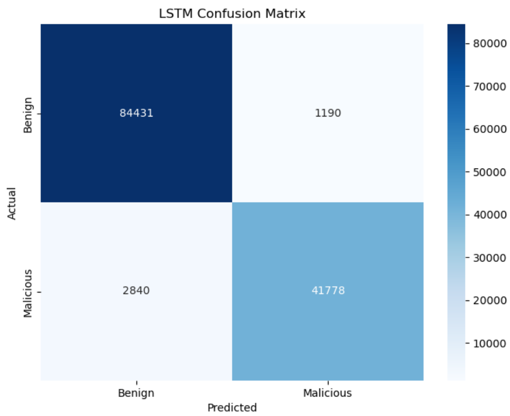
Table 2: LSTM Evaluation Metrics



(a) Accuracy Curves



(b) Loss Curves



(c) Confusion Matrix

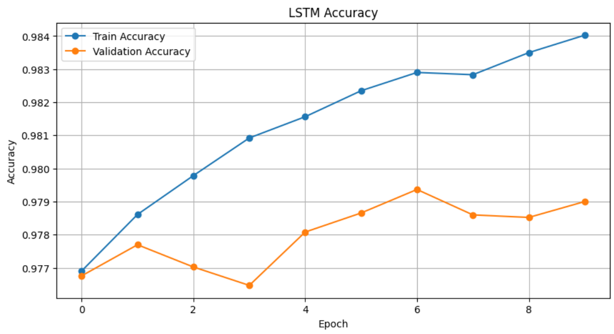
Figure 5: LSTM Performance Visualizations

0.4.3 LSTM with TF-IDF

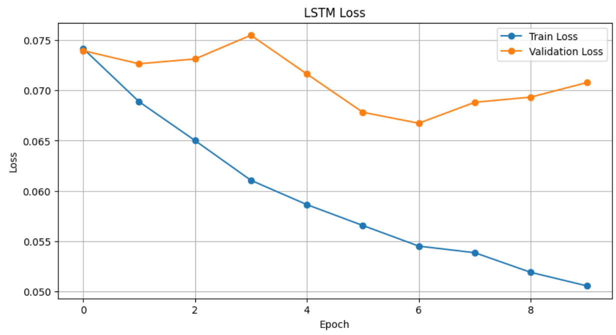
The LSTM with TF-IDF model outperformed others, achieving an overall accuracy of 0.98 (Table 3). It excelled in classifying benign and defacement URLs (F1-scores of 0.99) and performed robustly for malware and phishing URLs. The accuracy and loss curves (Figure 6) show rapid convergence, and the confusion matrix confirms minimal misclassifications. This model's ability to leverage textual patterns via TF-IDF features made it the optimal choice for deployment in the Gradio application.

Class	Precision	Recall	F1-Score	Support
Benign	0.98	0.99	0.99	85621
Defacement	0.99	1.00	0.99	19292
Malware	0.99	0.95	0.97	6504
Phishing	0.96	0.91	0.93	18822
Accuracy		0.98		130239
Macro Avg	0.98	0.96	0.97	130239
Weighted Avg	0.98	0.98	0.98	130239

Table 3: LSTM with TF-IDF Classification Report



(a) Accuracy Curves



(b) Loss Curves

Figure 6: LSTM with TF-IDF Performance Visualizations

0.5 Conclusion

The LSTM with TF-IDF model, leveraging textual patterns in URLs, achieved the highest accuracy (0.98) and was selected for deployment in a Gradio-based web application. This model provides robust real-time malicious URL detection, balancing performance and computational efficiency. Logistic Regression and LSTM models also performed well, offering interpretable and deep learning alternatives, respectively. The project demonstrates the efficacy of combining NLP and deep learning for cybersecurity applications. Future work includes:

- Exploring advanced NLP models like DistilBERT for lightweight embeddings.
- Implementing ensemble methods to combine Logistic Regression and LSTM predictions.
- Integrating real-time web scraping to enhance feature extraction with contextual data.
- Optimizing the Gradio interface for mobile compatibility and scalability.

0.6 References

- Malicious URLs dataset, Kaggle.
- `scikit-learn`, `tensorflow`, `urllib`, `tld`, `gradio` documentation.